

No. 25-13631

**UNITED STATES COURT OF APPEALS
FOR THE ELEVENTH CIRCUIT**

CITADEL SECURITIES, LLC,

Petitioner,

v.

SECURITIES AND EXCHANGE COMMISSION,

Respondent.

*On Petition for Review of an Order of the
Securities and Exchange Commission*

**BRIEF OF AMICUS LAW PROFESSOR J.W. VERRET
IN SUPPORT OF RESPONDENT**

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CERTIFICATE OF INTERESTED PERSONS

Pursuant to Federal Rule of Appellate Procedure 26.1 and 11th Circuit Rule 26.1-1, *amicus curiae* Professor J.W. Verret (“Verret”) discloses the names of all trial judges, attorneys, persons, associations of persons, firms, partnerships, or corporations that have an interest in the outcome of the particular case or appeal, including subsidiaries, conglomerates, affiliates, parent corporations, any publicly held corporation that owns 10% or more of the party's stock, and other identifiable legal entities related to a party:

1. Citadel Securities GP LLC, indirect, ultimate controlling entity of Citadel Securities LLC.
2. Citadel Securities LLC, Petitioner.
3. Keith Bradley and Squire Patton Boggs (US) LLP, counsel to amicus J.W. Verret.
3. Gibson, Dunn & Crutcher LLP, counsel for Petitioner Citadel Securities LLC.
4. Investors Exchange LLC, Intervenor.
5. Securities and Exchange Commission, Respondent.
6. Verret, J.W., Amicus Curiae in support of Respondent.

7. Wachtell, Lipton, Rosen & Katz, counsel for Intervenor Investors
Exchange LLC.

CORPORATE DISCLOSURE STATEMENT

Pursuant to Fed. R. App. P. 26.1 and 11th Cir. R. 26.1-1, *amicus curiae* submits the following disclosure statement: *Amicus curiae* is not a corporation, association, joint venture, partnership, syndicate, or other similar entity.

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INTEREST OF AMICUS CURIAE

Professor J.W. Verret is an Associate Professor of Law at George Mason University's Antonin Scalia Law School, where he teaches courses in securities regulation, corporate governance, and financial market structure. He has served as Chairman of the Market Structure Subcommittee of the SEC Investor Advisory Committee, where he led investigations into questions of equity market structure, exchange competition, and the effects of high-frequency trading on retail investors.¹ He has published extensively on issues of market structure and securities regulation, and he submitted a comment letter in the administrative proceedings below, supporting the SEC's approval of IEX's Options Facility.

Amicus files this brief to assist the Court in understanding the market structure context of this dispute, the role of market makers and risk controls in options markets, and the ways in which Citadel's characterizations of both the challenged mechanism and controlling precedent depart from the administrative record.

¹ No counsel for a party authored this brief in whole or in part, and no person other than *amicus* made a monetary contribution to the preparation or submission of this brief. All parties have consented to the filing of this brief.

SUMMARY OF ARGUMENT

Citadel’s challenge to the SEC’s approval of IEX’s Options Facility rests on a series of mischaracterizations that, upon examination, collapse under the weight of the administrative record and Citadel's own prior positions. Throughout its briefing, Citadel claims that a documented market phenomenon does not exist, that industry-standard practices are somehow unprecedented, and that a transparent, formula-driven tool to manage risk operates in ways it demonstrably does not.

To understand this dispute, the Court must first understand how options markets work. Options markets are quote-driven markets that depend entirely on market makers to provide displayed liquidity.² Unlike equity markets where retail and institutional investors regularly post limit orders, options markets feature a combinatorial explosion of contracts, with each underlying security potentially having hundreds of strike prices and expiration dates, that makes organic liquidity provision impractical. The large number of listed options renders the options

² U.S. Sec. & Exch. Comm’n, *Staff Report on Equity and Options Market Structure Conditions in Early 2021*, at 4 (Oct. 14, 2021), <https://www.sec.gov/files/staff-report-equity-options-market-struction-conditions-early-2021.pdf> (explaining that in options markets, “displayed liquidity is primarily derived from market maker quotes”) (“*Market Structure Report*”).

market dependent on market makers to provide liquidity. Market makers can account for over 99% of all open orders and quotes in options markets.³

Market makers, who may be responsible for hundreds of quotes across dozens of series in each class they quote, are especially vulnerable to latency arbitrage as underlying security prices move. As the SEC and various commenters have explained, "[i]n those moments when the price of an underlying equity security has change[d], a race condition exists between a market maker that needs to update and reprice its hundreds (and potentially thousands) of quotes across all strikes and expirations on that underlying security, and the liquidity takers attempting to trade with the market maker while its quotes remain at stale prices."⁴

Citadel's brief ignores this market reality. Instead, Citadel attempts to dismiss latency arbitrage as a fabricated concern when it is a well-known and documented problem to which market makers are acutely exposed in options markets. Citadel then attempts to thwart IEX from providing options trading by

³ Sandor Lehoczky et al., *Dead Man's Switch: Making Options Markets Safer with Active Quote Protection*, at 2 (May 2020), https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3675849 ('Options markets depend especially on market makers—who account for 99.9% of open orders—to connect buyers and sellers, due to a combinatorial explosion of expirations and strike prices.').

⁴ Order Approving a Proposed Rule Change to Adopt Rules Governing the IEX Options Facility, Exchange Act Release No. 34-103998, 90 Fed. Reg. 45,861, 45,870 (Sept. 23, 2025) ("*IEX Approval Order*").

mischaracterizing IEX's narrowly tailored risk tool, the Options Risk Parameter (ORP), which protects market makers from the predatory trading practices performed by latency arbitrageurs like Citadel and is similar to other exchanges' risk controls.

Citadel is asking the Court to defend Citadel's commercial interest but is ignoring these two principal issues that are known to the SEC and the industry at large: 1) latency arbitrage exists and discourages market makers from displaying valuable liquidity in the options market, and 2) there is significant and well documented precedent for exchanges that already offer protections to market makers to encourage liquidity provision just like the ORP.

The ORP is like the many risk controls other exchanges offer that allow market makers to manage their exposure to rapid price movements. "Options markets are characterized by ubiquitous repricing and canceling, with market makers employing sophisticated risk controls precisely because the alternative---maintaining stale quotes in rapidly moving markets---would be economically ruinous."⁵ Activity-based risk controls and other mechanisms have long permitted exchanges to cancel market maker quotes automatically based on pre-determined triggers.

⁵ *Id.* at 45,870, *citing* J.W. Verret, Comment Letter Re: Securities Exchange Act Release No. 34-102895 (SR-IEEX-2025-02), IEX Options Application, available at <https://www.sec.gov/comments/sr-iex-2025-02/sriex202502-616487-1808494.pdf>

The Court should reject Citadel's challenge in its entirety. Citadel is a primary beneficiary of the status quo. Citadel's challenge seeks to preserve its competitive advantages by blocking market innovations that would level the playing field, enhance market to market competition, and benefit retail investors through tighter spreads and deeper liquidity.

The SEC's approval was supported by substantial evidence, consistent with the Exchange Act's requirements, and squarely within the agency's expertise. Moreover, Citadel's core arguments have already been rejected by the D.C. Circuit in a prior challenge to IEX's D-Limit order type, a mechanism built on the same approved technologies and addressing the same market structure problem.⁶

ARGUMENT

I. OPTIONS MARKETS DEPEND ON MARKET MAKERS

Options markets are similar to equities markets but differ in their

⁶ In *Citadel Securities LLC v. SEC*, 45 F.4th 27 (D.C. Cir. 2022), the D.C. Circuit upheld the SEC's approval of IEX's D-Limit order, an innovation built on the same underlying technology and designed to address, in equities markets, the same market structure problem that the ORP addresses for options. Citadel raised substantially identical objections in that case: that IEX's mechanism was unfairly discriminatory against liquidity takers, that the SEC lacked substantial evidence of latency arbitrage, and that the mechanism undermined the protected quote rule. The D.C. Circuit rejected each of these arguments. The IEX ORP, by offering protection to market makers, provides a benefit to everyone in the market through tighter spreads and more liquidity, in the same way that the D.C. Circuit found the D-Limit tool provided a benefit to the market.

dependence on professional market makers. While equity markets benefit from diverse sources of displayed liquidity, options markets are quote-driven markets where professional market makers provide most of all displayed quotations.⁷

Understanding this structural feature is essential to evaluating Citadel's claims.

Moreover, in options each underlying security can have dozens of associated options contracts, spanning multiple strike prices and expiration dates. This vast number of available contracts means that organic liquidity provision, where diverse market participants independently post orders, is impractical. Instead, options markets depend on designated market makers who commit to continuously quoting two-sided markets across this vast array of contracts.⁸

These continuous quoting obligations create substantial risks to market makers. An options market maker quoting options on a single stock may be maintaining “hundreds (and sometimes thousands) of quotes on options for an underlying security at any one time.”⁹ When the price of the underlying equity

⁷ *Market Structure Report*, at 4 (explaining that in options markets, “displayed liquidity is primarily derived from market maker quotes”).

⁸ *Id.* See also *Dead Man's Switch: Making Options Markets Safer with Active Quote Protection*, at 2 (May 2020), https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3675849 (discussing the “combinatorial explosion of expirations and strike prices”).

⁹ *IEX Approval Order*, 90 Fed. Reg. at 45,870 (noting options market makers may maintain “hundreds (and sometimes thousands) of quotes on options for an underlying security at any one time”).

moves meaningfully, all of these options quotes may become stale simultaneously, exposing the market maker to significant losses if faster traders can execute against the old prices before the market maker can update them.

Recognizing this structural vulnerability, the SEC has for decades approved exchange-offered risk controls designed to help market makers manage their exposure. As the SEC found, options exchanges have long offered “purge ports that provide an expedited way for market makers to ‘mass cancel’ their quotes in bulk across classes and series simultaneously as well as options risk mitigation functionality in place at most options exchanges, activity-based risk limits, arbitrage checks, and intrinsic value checks, that allow market makers to direct an exchange to cancel their quotes or reject orders based on various triggers.”¹⁰

II. LATENCY ARBITRAGE EXISTS IN OPTIONS MARKETS AND IS MORE ACUTE THAN IN EQUITIES

A central premise of Citadel’s brief is its attempt to cast doubt on whether latency arbitrage is a real market phenomenon or whether the SEC properly understood its operation. Latency arbitrage is an extensively documented phenomena in modern market structure. It has been studied by economists, acknowledged by regulators, and has repeatedly been litigated in the DC Circuit.

¹⁰ *Id.* at 45,870-71.

Latency arbitrage exists because information travels at a finite speed. Securities exchanges are located in geographically dispersed data centers.¹¹ Even at the speed of light, it takes measurable time, microseconds to milliseconds, for price information generated at one exchange to reach another. This creates an exploitable window: traders with faster connections, superior technology, and proximity advantages can observe price movements at one venue and trade at stale prices at another venue before slower participants can react.

The “arms race” for speed in securities markets is well-documented. As the D.C. Circuit recognized, “[d]uring that latency, certain high-frequency traders can take securities at old, stale prices—just before updated prices reach the exchanges—and then turn around and trade those securities at the newly updated national best bid or offer.”¹² IEX itself has documented that latency between and

¹¹ See Eric Budish et al., *The High-Frequency Trading Arms Race: Frequent Batch Auctions as a Market Design Response*, 130 Q.J. Econ. 1547, 1548-49 (2015), <https://academic.oup.com/qje/article/130/4/1547/1916146>; see also Brief of Amicus Curiae XTX Markets LLC in Support of Respondent, *Citadel Sec. LLC v. SEC*, No. 20-1424, pp. 6-7 (D.C. Cir. Apr. 12, 2021) (“Latency arbitrageurs earn profits entirely from microseconds- to milliseconds-long price differences across the geographically dispersed data centers where the various relevant exchanges are located.”); Brief of Amicus Curiae Bloomberg L.P. in Support of Respondent, *Intercontinental Exchange, Inc. v. SEC*, No. 20-1470, pp. 14-15 (D.C. Cir.) (explaining that certain stock exchanges sell physical proximity for communications access because latency arbitrage depends on speed-of-light travel over shorter distances).

¹² *Citadel*, 45 F.4th at 31 (“[D]uring that latency, certain high-frequency traders can take securities at old, stale prices—just before updated prices reach the

among data centers located in New Jersey can range up to several hundred microseconds, with additional latency introduced by processing on both sides of any transaction.¹³

Academic research has consistently documented the harms of latency arbitrage. Professors Baldauf and Mollner found that latency arbitrage “has unambiguously detrimental implications, both increasing the spread and reducing information acquisition.”¹⁴ Professors Menkveld and Zoican found that when latency arbitrageurs are faster than liquidity providers, as they nearly always are, the latency differential “c[an] hurt liquidity” and cause liquidity providers “to set a wider [bid-ask] spread to recoup the increased adverse-selection cost.”¹⁵

Critically, recent peer-reviewed academic research has documented the specific impact of latency arbitrage on options markets. A 2024 study published in

exchanges—and then turn around and trade those securities at the newly updated national best bid or offer.”).

¹³ *IEX Approval Order*, 90 Fed. Reg. at 45,870 (stating that “latency between and among the data centers located in New Jersey range up to several hundred microseconds, with additional latency introduced by technology processing on both sides of an order or quote route between these data centers”).

¹⁴ *IEX Approval Order*, 90 Fed. Reg. at 45,870 (stating that “latency between and among the data centers located in New Jersey range up to several hundred microseconds, with additional latency introduced by technology processing on both sides of an order or quote route between these data centers”).

¹⁵ Albert Menkveld & Marius Zoican, *Need for Speed? Exchange Latency and Liquidity*, 30 Rev. Fin. Stud. 1188, 1214-15 (2017); *see also* Thierry Foucault et al., *Toxic Arbitrage*, 30 Rev. Fin. Stud. 1053, 1090 (2017) (finding that latency arbitrage leads to less liquidity and wider bid-ask spreads).

the Journal of Financial Economics found that “HFT activity in the equity markets is associated with a decline in market liquidity, as indicated by an increase in bid-ask spreads—in the options markets.”¹⁶ The authors specifically attributed this finding to the “latency arbitrage channel,” explaining that “option market makers’ quotes being exposed to sniping risk originating from latency races involving arbitrageurs who seek to profit from trading against stale quotes.”¹⁷ This research directly validates the SEC’s findings regarding latency arbitrage in options markets and confirms that the problem is “exclusively driven by aggressive HFT strategies and not by liquidity-supplying HFT strategies.”¹⁸

¹⁶ Mahendrarajah Nimalendran, Khaladdin Rzayev & Satchit Sagade, *High-Frequency Trading in the Stock Market and the Costs of Options Market Making*, 159 J. Fin. Econ. 103900 (2024), <https://www.sciencedirect.com/science/article/pii/S0304405X24001235> (“This association is exclusively driven by aggressive HFT strategies and not by liquidity-supplying HFT strategies.”).

¹⁷ Nimalendran, Rzayev & Sagade, *High-Frequency Trading in the Stock Market and the Costs of Options Market Making*, 159 J. Fin. Econ. 103900 (2024) (“[O]ur results are consistent with option market makers’ quotes being exposed to sniping risk originating from latency races involving arbitrageurs who seek to profit from trading against stale quotes—we call this explanation the *latency arbitrage* channel.”).

¹⁸ *Id.* (citing Eric Budish et al., *The High-Frequency Trading Arms Race*, 130 Q.J. Econ. 1547 (2015), and explaining that “latency arbitrage opportunities are flaws in market design, leading to increased trading costs”; concluding that “[w]e find a negative relationship between aggressive HFT strategies in the stock market and options market liquidity: more aggressive HFT activity in the stock market leads to wider bid-ask spreads in the options market”).

In options markets, the problem is even more acute than in equities. When the price of the underlying security moves, all options on that security must be repriced. A market maker quoting across dozens of strike prices and expirations faces the challenge of updating hundreds of quotes simultaneously. This creates a much larger window of vulnerability than in single-asset equity markets.

The administrative record documents these harms with specificity. One commenter, a former U.S. equity options market making firm, explained that it was “ultimately forced to exit the business due to the escalating costs and arms race associated with maintaining latency competitiveness.”¹⁹ The commenter observed that “[f]irms with deep expertise in option pricing and risk management—those best positioned to provide tight spreads and meaningful size—are often pushed out due to the steep technical demands required merely to remain competitive.”²⁰

The SEC credited this evidence, finding that “the potential for major losses resulting from latency arbitrage can cause options market makers to be less willing to quote their best possible price in the largest number of contracts they might otherwise display and that aversion to loss and inability to compete with the most technologically sophisticated firms can lead to market makers decreasing the

¹⁹ Letter from Brian P. Donnelly, Founder, Volant Trading, to Vanessa Countryman, Sec’y, SEC, pp. 1-2 (Aug. 8, 2025) (“*Volant Letter*”).

²⁰ *Id.* at 1.

number of options classes that they quote or leaving the business entirely.”²¹

These findings are consistent with academic research quantifying latency arbitrage across global markets. Professors Aquilina, Budish, and O'Neill analyzed London Stock Exchange message data and found that latency arbitrage, which they describe as “sniping” or “picking off” stale quotes, accounts for a substantial portion of trading activity, concentrated among a handful of firms trading during fractions of a percent of the trading day.²²

The D.C. Circuit has already upheld the SEC’s findings on latency arbitrage against Citadel’s prior challenges. In reviewing the D-Limit approval, the court found that the SEC had substantial evidence supporting the existence and harms of latency arbitrage, holding that “even though the speed bump helps liquidity providers more than liquidity removers, there is no unfair discrimination because the order type ultimately makes all market participants better off by improving overall liquidity.”²³ Citadel now asks this Court to reach a contrary conclusion on substantially similar facts.

²¹ *IEX Approval Order*, 90 Fed. Reg. at 45,878-79.

²² Matteo Aquilina et al., *Quantifying the High-Frequency Trading “Arms Race”*, 137 Q.J. Econ. 493, 495-96, 520 (2022), <https://academic.oup.com/qje/article/137/1/493/6368348> (finding that such activity totals approximately 0.043 seconds per day, or about 0.0001% of the trading day, concentrated among a handful of firms).

²³ *Citadel*, 45 F.4th at 35.

III. CITADEL'S CLAIM THAT IEX QUOTES WILL BE “ILLUSORY” IS FALSE AND IGNORES THAT CITADEL AND OTHER MARKET MAKERS RELY ON RISK CONTROLS SIMILAR TO THE IEX ORP

Citadel’s brief deploys the pejorative term “illusory” to mischaracterize IEX’s ORP, incorrectly suggesting that IEX has created some unprecedented mechanism that allows market makers to back away from their obligations or to somehow suggest that their quotes are not “firm.” This characterization is false.

First, as the SEC explained, “[t]he ability of any market participant to successfully execute against any particular displayed quote is subject to a number of factors and is not guaranteed on any market, as at any time any market participant can be seeking to execute against an order that is being repriced, changed, cancelled, or executed by a different market participant.”²⁴ Second, it ignores that every options exchange offers risk controls that permit automated quote cancellation, mechanisms that Citadel itself uses and has publicly defended.²⁵ Third, it contradicts Citadel’s own published research explaining why

²⁴ See <https://www.sec.gov/files/rules/sro/iex/2025/34-103998.pdf> at 28-29.

²⁵ See “Options exchanges commonly offer optional risk mitigation functionality (also called risk controls) that allow market makers and others to have the exchange automatically cancel their quotes and orders when certain triggers specified by the market participant are met.” <https://www.sec.gov/files/rules/sro/iex/2025/34-103998.pdf> at page 23.

rapid quote updating and cancellation is not only normal but essential to healthy market structure.

Citadel has published white papers explaining in detail why its high rates of frequent quote cancellation and updating to reflect market conditions are beneficial to markets. In its July 2020 “Market Lens” paper, Citadel explained that “order cancellations allow automated traders to dynamically adjust their prices to rapid changes in supply and demand, which results in tighter spreads and **better execution for all market participants.**”²⁶ The paper further observed that “rapidly changing price quotations and order cancellations” have “become an integral part of our market structure that reduce volatility, tighten spreads and lower transaction costs.”²⁷ The paper concluded that “high quote cancellation rates have become not only normal, but also integral to the proper functioning of modern markets, resulting in greater efficiency, narrower bid-ask spreads, and more robust price discovery.”²⁸

Citadel doubled down on this position in a second “Market Lens” white paper published in October 2021, which provided “additional evidence on the

²⁶ Citadel Securities, *Market Lens: Why Do Electronic Traders Cancel Orders?*, at 1 (July 2020), https://www.citadelsecurities.com/wp-content/uploads/sites/2/2020/07/Market-Lens-Order-Cancellation-White-Paper_FINAL.pdf.

²⁷ *Id.*

²⁸ *Id.* at 3.

inextricable relationship between bid-ask spreads and cancel rates, why high levels of order cancellations and constant price updating fit into a healthy market structure to allow for risk management, and how this ultimately benefits investors.”²⁹ Yet when IEX offers a transparent, narrowly tailored rule-based mechanism to protect market makers from undue risk, Citadel feigns surprise that quotes can be canceled, suggesting that a routine and natural part of modern markets somehow transforms into “illusory quotes” or “unlawful backing away” when it’s not Citadel initiating the cancellations.

Major market-making firms have similarly explained the importance of risk controls in options markets. Optiver, another leading market maker, has published research explaining that liquidity providers maintain “hundreds of quotes on a given underlying at any one time” and that sudden market movements leave them “vulnerable to stale quotes.”³⁰ Optiver has noted that market-maker protections “are one of the most frequently used forms of liquidity protection in the options market” and “give market makers the confidence to remain in the market and

²⁹ Citadel Securities, *Market Lens: Why Restricting Cancel Rates Can Increase Bid-Ask Spreads* (Oct. 2021), <https://www.citadelsecurities.com/wp-content/uploads/sites/2/2021/10/Why-Restricting-Cancel-Rates-Can-Increase-Bid-Ask-Spreads.pdf>

³⁰ Optiver, *Protecting Liquidity in Options Markets* (July 12, 2023), <https://optiver.com/insights/protecting-liquidity-in-options-markets/>.

provide continuous, high-quality liquidity in both normal and volatile conditions.”³¹

The administrative record demonstrates that every options exchange offers risk controls that can result in automated quote cancellation, and that these mechanisms have never been considered improper, nor have they been considered to cause “illusory quotes” or considered “non-firm” or “unprotected” quotes. As the SEC found, options exchanges have long offered “purge ports that provide an expedited way for market makers to ‘mass cancel’ their quotes in bulk across classes and series simultaneously as well as options risk mitigation functionality in place at most options exchanges—activity-based risk limits, arbitrage checks, and intrinsic value checks—that allow market makers to direct an exchange to cancel their quotes or reject orders based on various triggers.”³² These mechanisms “have never been viewed as allowing ‘quote fading’ by an exchange or the market makers that choose to use them.”³³

These mechanisms are ubiquitous across all options exchanges. NYSE’s options platforms state that “Activity-based risk limits are mandatory for

³¹Optiver, *Market-Maker Protections* (July 17, 2023), <https://optiver.com/insights/market-maker-protections/>.

³² *IEX Approval Order*, 90 Fed. Reg. at 45,871.

³³ *Id.* at 45,870.

order/quote entry in all underlying symbols” for options market makers.³⁴ Because Citadel is a leading U.S. options market maker, this means Citadel itself is required to use activity-based risk controls.^{35,36} MX2 Options (MEMX’s recently approved *second options venue*)³⁷ offers “pre-trade risk controls, activity-based risk controls, and global risk controls” that are “designed to offer protection from entering orders outside of certain size and price parameters.”³⁸

Nasdaq PHLX offers a Risk Monitor Mechanism that allows market makers to automatically remove their quotes based on configurable parameters, explaining that “[w]ithout adequate risk management tools in place on the Exchange, the

³⁴ NYSE Group, *NYSE Pillar Risk Controls*, § 6.9, at 17 (Jan. 15, 2026), https://www.nyse.com/publicdocs/nyse/NYSE_Pillar_Risk_Controls.pdf.

³⁵ *Id.*

³⁶ See Letter from Adam C. Cooper, Chief Legal Officer, Citadel Securities, to Brent J. Fields, Sec’y, SEC, at 1 n.1 (Jan. 17, 2017), <https://www.sec.gov/comments/sr-chx-2016-16/chx201616-1504185-130596.pdf> (stating Citadel “accounts for . . . 19 percent of U.S. listed equity option volume”).

³⁷ Citadel is a significant investor in MEMX and provided funding to MEMX to launch its options exchange. See MEMX expands offering with launch of US options exchange-The TRADE. Available at <https://www.thetradenews.com/memx-expands-offering-with-launch-of-us-options-exchange/#:~:text=%E2%80%9CI%20am%20proud%20and%20grateful,Citadel%20Securities%20and%20Virtu%20Financial>. See also MEMX LLC Agreement (naming Citadel as a significant investors and market maker member of MEMX), available at <https://info.memxtrading.com/wp-content/uploads/2025/10/MEMX-Holdings-LLC-8th-AR-LLC-Agreement-10.9.25.pdf>.

³⁸ Order Approving a Proposed Rule Change to Establish MX2 Options, Exchange Act Release No. 34-104152, pp. 19-22, 37 (Sept. 30, 2025), <https://www.sec.gov/files/rules/sro/mx2/2025/34-104152.pdf>.

incentive for Market Makers to quote aggressively, respecting both price and size could be diminished.”³⁹ Similarly, MIAX has offered its Aggregate Risk Manager (ARM) for market makers since 2012 that allows market makers to set their own customizable, non-public risk setting parameters.⁴⁰ NYSE Arca has activity-based controls under its Rule 6.40P-O, as do other options exchanges.⁴¹ Activity-based risk controls operate automatically based on parameters set in advance and known only to the market maker and the exchange. When triggered, they can result in the

³⁹ Self-Regulatory Organizations; NASDAQ PHLX LLC; Notice of Filing and Immediate Effectiveness of Proposed Rule Change Relating to the Risk Monitor Mechanism, Exchange Act Release No. 34-78129, 81 Fed. Reg. 41,710 (June 28, 2016), <https://www.govinfo.gov/content/pkg/FR-2016-06-28/pdf/2016-15177.pdf>.

⁴⁰ MIAX Options Exchange Rulebook, Rule 612 (Aggregate Risk Manager), https://www.miaxglobal.com/miax_options_exchange_rules.pdf (adopted Dec. 3, 2012); *see also* Self-Regulatory Organizations; MIAX; Notice of Filing and Immediate Effectiveness of a Proposed Rule Change to Amend Exchange Rule 612, Aggregate Risk Manager (“ARM”), Exchange Act Release No. 34-77817 (May 12, 2016), <https://www.sec.gov/files/rules/sro/miax/2016/34-77817.pdf>. (stating that the market maker risk tools “protects investors and the public interest by ensuring that liquidity providers such as Exchange Market Makers are able to quote aggressively within their risk tolerance levels with respect to both price and size, resulting in narrower bid/ask differentials and deeper liquidity on the Exchange, all to the benefit and protection of investors and the public interest”).

⁴¹ *See* Self-Regulatory Organizations; NYSE Arca, Inc.; Notice of Filing of Amendment No. 4 and Order Granting Accelerated Approval of a Proposed Rule Change to Adopt New Rules, Exchange Act Release No. 34-94072 (Jan. 26, 2022), <https://www.sec.gov/files/rules/sro/nysearca/2022/34-94072.pdf> (approving Rule 6.40P-O for NYSE Arca options). It is worth noting that market-maker risk controls have existed for many years as consistent with the Exchange Act, and that, as reflected in the SEC-approved orders noted above, no exchange seeking approval offered evidence approaching the depth and substance of the record IEX provided to the SEC.

exchange canceling all a market maker's quotes across an entire class of options. Critically, these cancellations happen "on their own initiative" by the exchange, "with no explicit, contemporaneous instruction" from the market maker.⁴² These mechanisms "have never been viewed as allowing 'quote fading' by an exchange or the market makers that choose to use them."⁴³

And yet these mechanisms are also fundamentally indistinguishable from the IEX ORP that Citadel challenges in this case. IEX's ORP is simply a more transparent version of this established functionality that is narrowly tailored (estimated to affect quotes less than 1% of the trading day on average) and designed to protect options market makers from undue risks including latency arbitrage.

The omission of activity-based controls from Citadel's brief is telling. Citadel is a dominant market maker that benefits from these controls on every exchange where it operates. For example, on NYSE Arca options platform, Citadel is appointed as a lead market maker in over 40% of listed options classes, and NYSE requires all market makers to use activity-based risk controls.⁴⁴

⁴² *CTC Letter*, at 4.

⁴³ *IEX Approval Order*, 90 Fed. Reg. at 45,870.

⁴⁴ See website of NYSE Arca, NYSE Options, list of lead market makers publicized, available at <https://www.nyse.com/trade/options>. See list of lead market makers publicized at <https://www.nyse.com/publicdocs/nyse/markets/arca-options/EligibleOptionsArca.xls.txt>. And see also

The ORP's innovation is not that it allows exchange-mediated quote cancellation, that has existed and currently exists on exchanges,⁴⁵ but that it does so in a more transparent, narrowly targeted way that will enhance market maker competition and benefit all market participants.⁴⁶ Citadel's characterization of the ORP as a "pause-and-peek" mechanism that allows market makers to back away from displayed quotes fundamentally misrepresents how the mechanism actually operates. The SEC expressly found that the ORP "will not offer a 'last look' to a Market Maker by allowing it to back away from a quote when presented with an incoming order."⁴⁷

The mechanism works as follows. When a market maker submits a quote, it may elect voluntarily in advance to have the ORP apply to that quote. If selected, the ORP uses a predetermined formula, the "Indicator," to assess whether in a

https://www.nyse.com/publicdocs/nyse/NYSE_Pillar_Risk_Controls.pdf (activity-based controls are "mandatory for Options Market Makers"). Tellingly, Citadel fails to mention or disclose to the Court its highly concentrated market making positions on other options exchanges.

⁴⁵ See <https://www.sec.gov/files/rules/sro/iex/2025/34-103998.pdf>, "Updating or cancelling options quotes in response to a change in the price of the underlying equity security is part of the price discovery process for options and activity that a market maker is constantly doing as part of its continuous quoting requirement.").

⁴⁶ See <https://www.sec.gov/files/rules/sro/iex/2025/34-103998.pdf>, ("[B]y protecting Market Makers in this narrowly tailored way, IEX may attract additional liquidity, including from new market makers, which will promote more displayed liquidity that will be available to all market participants.").

⁴⁷ *IEX Approval Order*, 90 Fed. Reg. at 45,871.

particular options series the price is sufficiently dislocated from the price of the underlying security to indicate that the options series is in transition and the quote is likely stale. The ORP formula uses the Black-Scholes options pricing model, which is the foundational mathematical framework for pricing options contracts. Developed by Fischer Black and Myron Scholes in 1973, the model calculates the theoretical value of an option based on several inputs: the current price of the underlying asset, the option's strike price, the time to expiration, the risk-free interest rate, and the volatility of the underlying asset. The key insight of Black-Scholes is that an option's price can be derived by constructing a risk-free hedge between the option and its underlying security.⁴⁸ The model has become the industry standard for options pricing and is widely used by exchanges, market makers, and investors to determine fair option values. If the market maker's quote is above (for a bid) or below (for an offer) the price level calculated by the Indicator, IEX will effectuate the market maker's *preexisting instructions* to either cancel or reprice the quote to the calculated price.⁴⁹ The key parameters of this formula are set transparently in IEX's rules.⁵⁰

⁴⁸ See Fischer Black & Myron Scholes, *The Pricing of Options and Corporate Liabilities*, 81 J. Pol. Econ. 637 (1973); see also Investopedia, *Black-Scholes Model: What It Is, How It Works, and the Options Formula*, <https://www.investopedia.com/terms/b/blackscholes.asp>.

⁴⁹ *IEX Approval Order*, 90 Fed. Reg. at 45,866-67, 45,869.

⁵⁰ *Id.* at 45,872.

The Court should not entertain Citadel’s attempt to recast settled and widely accepted market practices, which benefit all market participants, as something suspect or unprecedented.

CONCLUSION

The SEC’s approval of IEX's Options Facility reflects careful, reasoned decision-making supported by substantial evidence and consistent with binding precedent. Citadel’s challenge fails at every turn.

Options markets depend on market makers, and exchanges have long offered risk controls to support market maker function. The ORP is a more transparent and narrowly tailored version of functionality that has existed for decades. Latency arbitrage is a documented phenomenon that imposes significant costs on options markets. Multiple market participants with direct experience, including market makers who have been forced to exit the business due to the costs of defending against latency arbitrage, provided detailed evidence that the SEC reasonably credited.

Citadel’s characterization of “illusory” quotations ignores that every options exchange offers risk controls permitting automated quote cancellation, mechanisms that Citadel use daily and mechanisms that are consistent with its own research. The ORP is more transparent and more narrowly targeted than existing alternatives, not less. And Citadel’s description of the ORP as a “pause-and-peek”

mechanism is simply inaccurate. The ORP operates automatically based on a fixed, publicly disclosed formula. Market makers have no ability to selectively back away from orders.

At bottom, Citadel asks this Court (after Citadel lost before the D.C. Circuit on an uncannily similar challenge) to block a pro-competitive innovation that would reduce barriers to entry, encourage tighter spreads, and benefit investors, all to protect Citadel's dominant position in a concentrated market. The Exchange Act does not require the SEC to preserve incumbent advantages at the expense of market quality and the public interest.

The petition for review should be denied.

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Respectfully submitted,

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